

AMENDMENT NO. 1 OCTOBER 2022

TO

**IS 16481 : 2022 TEXTILES — SYNTHETIC MICRO-FIBRES FOR USE IN CEMENT
BASED MATRIX — SPECIFICATION**
(*First Revision*)

(Foreword, para 3, sentence 4) — Delete.

(Foreword, para 3, sentence 5) — Delete the word ‘temperature’.

(Foreword, para 4) — Insert the following after the paragraph:

‘Therefore, use of synthetic fibres such as polyester, nylon and polypropylene in construction needs no emphasis.’

(Foreword, para 5, sentence 1) — Delete.

(Foreword, para 5, sentence 2) — Delete ‘reduced seepage of water’.

(Foreword, para 5, sentence 3) — Delete.

[Page 5, Table 1, Sl No. v) col (3), (4) and (5)] — Substitute ‘240°C’ for ‘250°C’.

[Page 5, Table 1, Sl No. vi) col (3), (4) and (5)] — Substitute ‘60°C’ for ‘80°C’.

(Page 17, Annex G) — Substitute the following for the existing:

‘ANNEX G
[Sl No. iii), Table 1]

DETERMINATION OF SPECIFIC GRAVITY

G-1 Take a specimen of weighing about 1 g of fibres from the sample and find out its mass (m) accurately to the nearest mg. Count the total number of fibres (*n*) in the test specimen.

G-2 Let *l* be the mean fibre length, in cm, correct to two decimal places as determined by the procedure described in IS 235 and *d* be the mean fibre diameter in cm, correct to two decimal places as determined in IS 6919.

Price Group 1

G-3 For circular cross-section fibres, calculate the specific gravity of the fibre by the formula:

$$\text{Specific gravity of the fibre} = \frac{4m}{\pi d^2 l n}$$

where

m = mass of specimen in mg;

l = length of fibre in cm;

d = mean fibre diameter in cm; and

n = total number of fibres.

G-4 For triangular cross-section fibres, calculate the specific gravity of the fibre by the formula:

$$\text{Specific gravity of the fibre} = \frac{m}{a \times \sin 60 \times a/2 \times l \times n} = \frac{m}{0.433 a^2 \times l \times n}$$

Where

a = side of the equilateral triangle of cross-section of the fibre.

G-5 For rectangular cross-section fibres, calculate the specific gravity of the fibre by the formula:

$$\text{Specific gravity of the fibre} = \frac{m}{a \times b \times l \times n}$$

Where

a = longer side of the rectangle of cross-section of the fibre; and

b = shorter side of rectangle of cross-section of the fibre.'